Assignment 1 : Linux Basics

By - Ankita Sahoo

GitHub Repo Link:https://github.com/ankiitaasahoo19/Linux-Commands.git

1. **Creating and Renaming Files/Directories**
   Created a new directory named test_dir.
   Created an empty file called **example.txt** inside the **test_dir** directory.
   Renamed the file **example.txt** to **renamed_example.txt**.

```
ankitas@Ankita:~$ ls
ankitas@Ankita:~$ mkdir test_dir
ankitas@Ankita:~$ cd test_dir
ankitas@Ankita:~/test_dir$ touch example.txt
ankitas@Ankita:~/test_dir$ mv example.txt renamed_example.txt
ankitas@Ankita:~/test_dir$ ls
renamed_example.txt
```

2. **Viewing File Contents**
   Displays the entire contents of the **/etc/passwd** file.
   Shows the first 5 lines of the file.
   Shows the last 5 lines of the file.

```
ankitas@Ankita:~/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin
/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/s
bin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/fa
lse
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
ankitas:x:1000:1000:,,,:/home/ankitas:/bin/bash
```

```
ankitas@Ankita:~/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
ankitas@Ankita:~/test_dir$ tail -n 5 /etc/passwd
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
ankitas:x:1000:1000:,,,:/home/ankitas:/bin/bash
```

3. **Searching for Patterns**
   Searched and printed all lines in **/etc/passwd** that contain the word **root** using **grep**.

```
ankitas@Ankita:~/test_dir$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

4. **Zipping and Unzipping**
   Created an archive named **test_dir.zip** containing the **test_dir** folder using **tar -cvzf**.

```
ankitas@Ankita:~/test_dir$ cd ..
ankitas@Ankita:~$ tar -cvzf test_dir.zip test_dir
test_dir/
test_dir/renamed_example.txt
ankitas@Ankita:~$ ls
test_dir  test_dir.zip
```

   Created a directory named **unzipped_dir** to store extracted files.
   Extracted the contents of **test_dir.tar** into **unzipped_dir** using **tar -xvf**.

```
ankitas@Ankita:~$ mkdir unzipped_dir
ankitas@Ankita:~$ tar -xvf test_dir.zip -C unzipped_dir
test_dir/
test_dir/renamed_example.txt
ankitas@Ankita:~$ ls
test_dir  test_dir.zip  unzipped_dir
```

5. **Downloading Files**
   Downloaded a file from the internet and saved it in the current directory.

```
ankitas@Ankita:~$ wget https://pagertree.com/learn/assets/images
/devops-loop-46ccaf695c535cd2a81764d58414c6d7.webp
--2026-03-03 17:46:05--  https://pagertree.com/learn/assets/imag
es/devops-loop-46ccaf695c535cd2a81764d58414c6d7.webp
Resolving pagertree.com (pagertree.com)... 104.26.13.63, 172.67.
68.42, 104.26.12.63, ...
Connecting to pagertree.com (pagertree.com)|104.26.13.63|:443...
 connected.
HTTP request sent, awaiting response... 200 OK
Length: 28386 (28K) [image/webp]
Saving to: ‘devops-loop-46ccaf695c535cd2a81764d58414c6d7.webp’

devops-loop-46c 100%[=======>]  27.72K  --.-KB/s    in 0.002s

2026-03-03 17:46:06 (14.2 MB/s) - ‘devops-loop-46ccaf695c535cd2a
81764d58414c6d7.webp’ saved [28386/28386]

ankitas@Ankita:~$ ls
devops-loop-46ccaf695c535cd2a81764d58414c6d7.webp  test_dir.zip
test_dir                                           unzipped_dir
```

6. **Changing Permissions**
   Created a new empty file named **secure.txt**.
   Makes the file **read-only for all users** using **chmod 444 secure.txt**.

```
ankitas@Ankita:~$ touch secure.txt
ankitas@Ankita:~$ ls -l
total 40
-rw-r--r-- 1 ankitas ankitas 28386 Feb 19 17:19 devops-loop-46cc
af695c535cd2a81764d58414c6d7.webp
-rw-r--r-- 1 ankitas ankitas     0 Mar  3 17:47 secure.txt
drwxr-xr-x 2 ankitas ankitas  4096 Mar  3 17:18 test_dir
-rw-r--r-- 1 ankitas ankitas   167 Mar  3 17:32 test_dir.zip
drwxr-xr-x 3 ankitas ankitas  4096 Mar  3 17:40 unzipped_dir
ankitas@Ankita:~$ chmod 444 secure.txt
ankitas@Ankita:~$ ls -l
total 40
-rw-r--r-- 1 ankitas ankitas 28386 Feb 19 17:19 devops-loop-46cc
af695c535cd2a81764d58414c6d7.webp
-r--r--r-- 1 ankitas ankitas     0 Mar  3 17:47 secure.txt
drwxr-xr-x 2 ankitas ankitas  4096 Mar  3 17:18 test_dir
-rw-r--r-- 1 ankitas ankitas   167 Mar  3 17:32 test_dir.zip
drwxr-xr-x 3 ankitas ankitas  4096 Mar  3 17:40 unzipped_dir
```

7. **Working with Environment Variables**

   Creates an environment variable named **MY_VAR** with the value **Hello, Linux!**.

```
ankitas@Ankita:~$ export MY_VAR="Hello, Linux!"
ankitas@Ankita:~$ echo $MY_VAR
Hello, Linux!
```